

Maths Mock 1 — answer booklet

Answers and marking for Papers 1, 2 and 3

Award the marks shown. For 2-mark questions the right-hand column says when to give 1 mark for a correct method. Answers to fraction questions in Paper 1 may be given in any equivalent form unless the question asks for the simplest form.

Paper 1: arithmetic (40 marks)

Q	Answer	Marks	Notes
1	936	1	
2	229	1	
3	900	1	
4	55.523	1	
5	398	1	
6	7	1	
7	4,482	1	
8	1,710	1	
9	54	1	
10	8	1	
11	$\frac{4}{9}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
12	132	1	
13	8,631	1	
14	35,742	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
15	250	1	
16	28.5	1	
17	3,961,000	1	
18	$\frac{1}{3}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
19	35	1	

20	966,117	1	
21	1,624	1	
22	$1\frac{3}{8}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
23	24.2	1	
24	$7\frac{1}{2}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
25	36	1	
26	$\frac{2}{35}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
27	190	1	
28	501	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
29	$3\frac{1}{24}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
30	336,168	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
31	64	1	
32	1,495	1	
33	45	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
34	294	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
35	100	1	
36	$3\frac{4}{15}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.

Paper 2: reasoning (35 marks)

Q	Answer	Marks	Notes
1	C	1	
2	975 and 1,200	1	
3	£1.88	1	
4a	(7, 1)	1	
4b	(2, 9)	1	
5a	9	1	
5b	2,917,513	1	
6	64 and 112	1	
7a	5 miles	1	
7b	16 km	1	
8	isosceles triangle	1	
9a	9	1	
9b	5	1	
10	17:55	1	
11	35 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $197 - 52 = 110$
12	Explanation that 1 hour is 60 minutes, so 90 minutes is more than 1 hour (1 hour 30 minutes).	1	
13	$(30 \times 5) + (7 \times 5)$ and $(5 \times 40) - (5 \times 3)$	2	Award 2 marks for both correct calculations ticked and no others. Award 1 mark for one correct calculation ticked and no others, or both correct and one wrong.
14	39 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
15	52%	1	
16	0.25 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $2 \frac{1}{2} = 2.5$, then $2.5 - 2.25$

17	£0.45	1	
18	14/17, 15/19, 17/20	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
19	£42	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $1,335 \div 15 = 89$, then $89 - 33 - 14$
20	5 and 19; 7 and 17; 11 and 13	2	Award 2 marks for all 3 pairs and no incorrect pairs (in any order). Award 1 mark for 2 correct pairs and no incorrect pairs.
21	331 g	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. £1 is $50 \times 2\text{p}$ coins: $50 \times 7.12 = 356$ g; £1 is $5 \times 20\text{p}$ coins: $5 \times 5 = 25$ g; then subtract
22	108 cm ³	1	
23	14 km	1	

Paper 3: reasoning (35 marks)

Q	Answer	Marks	Notes
1a	1,044,000, 1,047,800	1	
1b	296,911, 576,094	1	
2	26,700	1	
3	£2.34	1	
4	(7, 2)	1	
5	3,075 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $13 \times 25 = 325$, then $2,750 + 325$
6	$29/6 = 4 \frac{5}{6}$; $13/6 = 2 \frac{1}{6}$; $37/6 = 6 \frac{1}{6}$; $17/6 = 2 \frac{5}{6}$	2	Award 2 marks for all four lines correct. Award 1 mark for two or three lines correct.
7	£3.80	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
8a	0.2	1	
8b	-7	1	
9	The giraffe is more than three times as tall as the hippo. The rhino is 50 cm taller than the hippo.	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
10	11:50	1	
11	7/1000, 0.15, 0.154, 0.75	1	
12a	$y = 12$	1	
12b	$y = 20$	1	
13a	4,560	1	
13b	77	1	
14	10	1	
15	30 days	3	Award 3 marks for the correct answer. Award 2 marks for an appropriate complete method with no more than one arithmetic error, e.g. $60 \times 24 = 1,440$, then $1,440 \div 48$. Award 1 mark for $60 \times 24 = 1,440$ seen, or for dividing a total by 48.
16a	6,450 m	1	

16b	0.175 m	1	
17	22 cm	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $38.5 \div 7 = 5.5$, then 5.5×4
18	$\frac{1}{4}$	1	Answer must be in its simplest form.
19	18, 81	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
20	66°	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $180 - 115 = 65$, then $180 - 49 - 65$
21	200 g	1	